

RC Controller Car Project Report

1. Abstract

This project presents an Arduino-based RC controller car that is operated wirelessly using a Bluetooth or RF controller. The system controls two DC motors through an L298N motor driver.

2. Introduction

An RC controller car is a mobile robotic platform used for learning embedded systems, motor control and wireless communication.

3. Objectives

- Build a wireless RC car.
- Learn Arduino programming.
- Control DC motors using L298N.
- Understand wireless communication.

4. Components Required

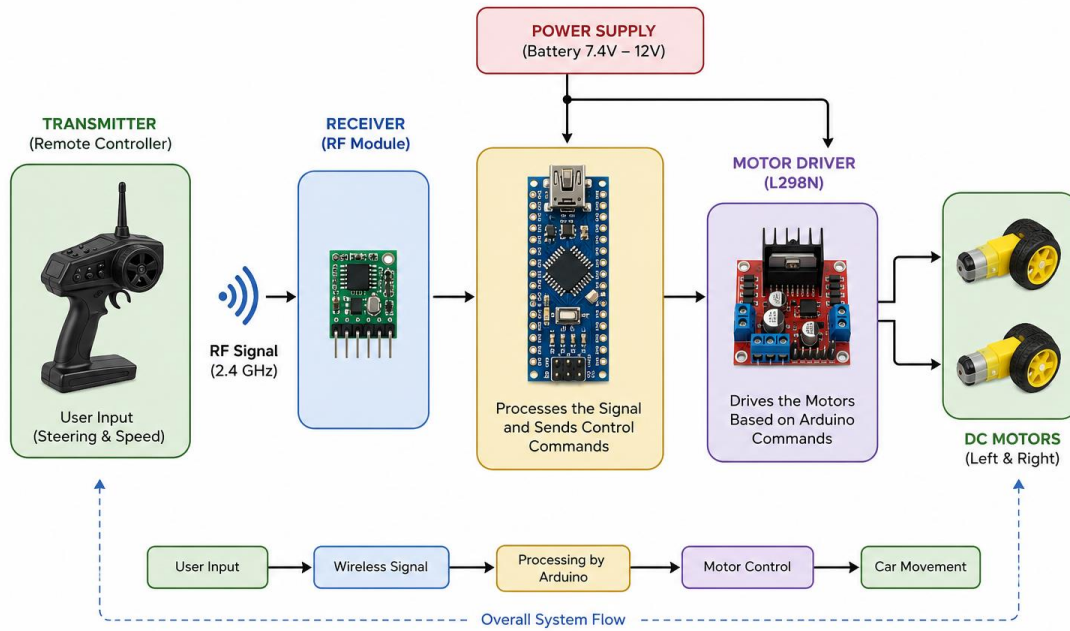
Arduino Uno
L298N Motor Driver
2 DC Gear Motors
Robot Chassis
Wheels
Caster Wheel
Battery
Bluetooth HC-05 (or RF Module)
Jumper Wires

5. Working Principle

The controller sends movement commands. Arduino receives the command and drives the left and right motors via the L298N motor driver to move forward, backward, left or right.

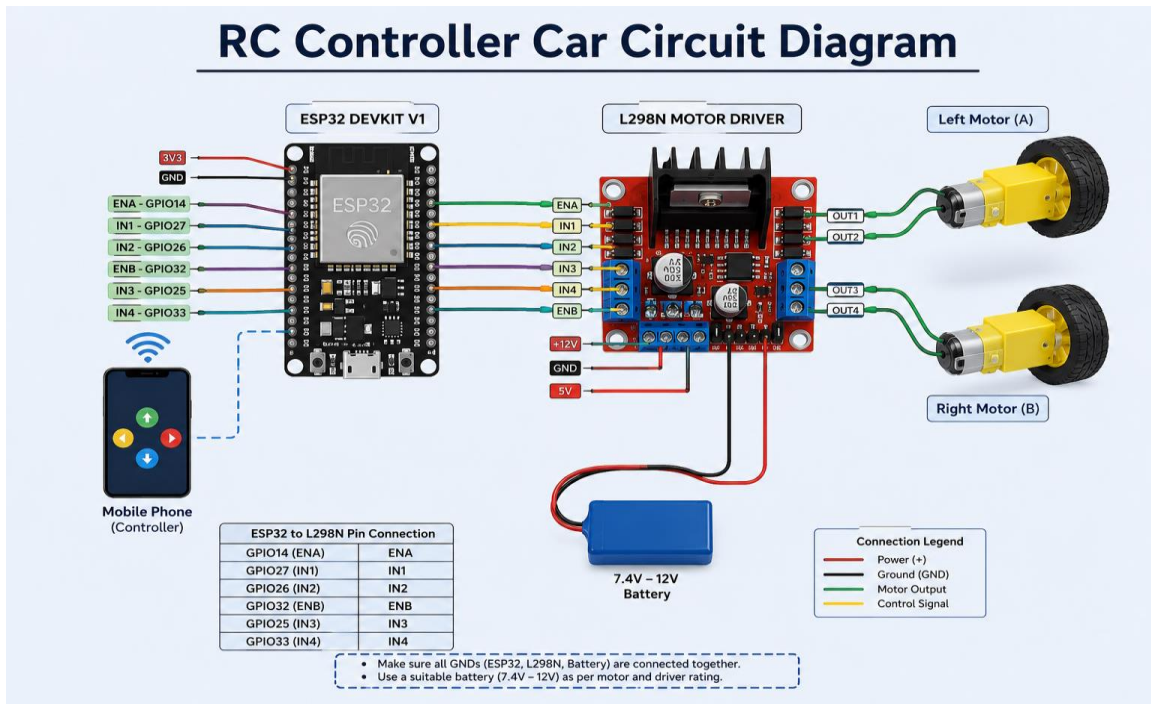
6. Block Diagram

RC Controller Car Block Diagram



Controller → Bluetooth/RF → Arduino → L298N → Motors]

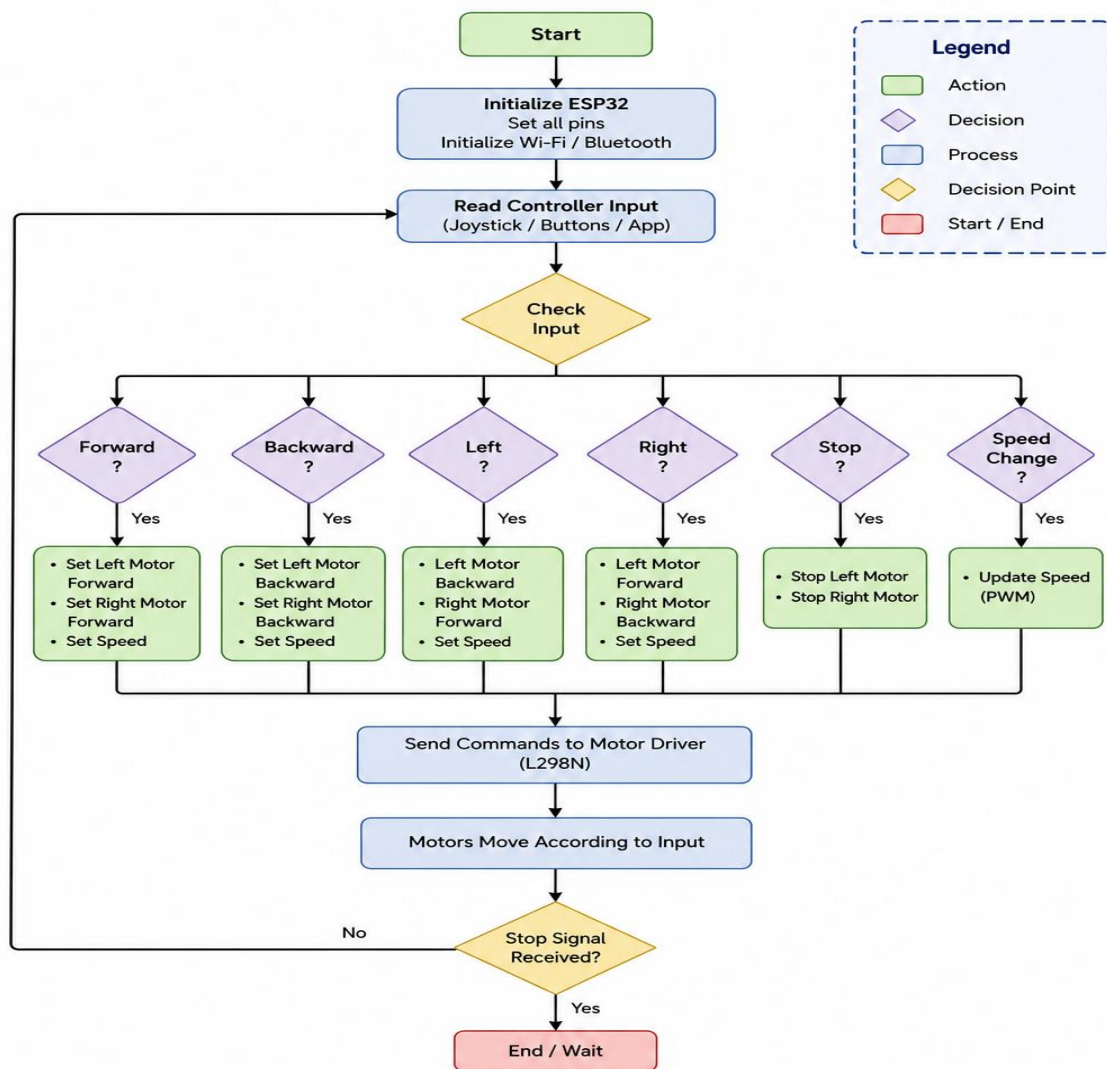
7. Circuit Diagram



Complete wiring diagram of Arduino, L298N, Motors, Battery and Bluetooth/RF Module

8. Flowchart

RC Controller Car Flowchart



Note: The car continuously reads input from the controller and sends appropriate commands to the motors for smooth and real-time movement.

Start → Initialize → Receive Command → Move Robot → Wait for Next Command

9. Advantages

Easy to build, low cost, educational, expandable and reusable.

10. Applications

Robotics education, surveillance prototype, warehouse robots, hobby projects.

11. Future Scope

Add obstacle avoidance, FPV camera, GPS, Wi-Fi/ESP32 control, autonomous navigation.

12. Conclusion

The RC Controller Car demonstrates wireless robotic control using Arduino and can be enhanced with sensors and AI features.

13. References

Arduino.cc

L298N datasheet

HC-05 documentation